LLM-Agnostic Multi-Provider Orchestration for Security AI Systems

*Note: Sub-title as needed

1st Nishan Paul
Research & Development
Joblio Security Labs
Dhaka, Bangladesh
nishan.paul.hello@gmail.com

2nd Collaborator Name
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

***Abstract*—Enterprise deployment of AI-powered security assessment systems faces a fundamental challenge: no single LLM provider is optimal for all security tasks. We present CMatrix-LLMOrch, a provider-agnostic LLM orchestration layer for security AI systems that provides a unified LLMProvider interface across six distinct provider backends. Each provider implements a consistent contract with automatic message format translation, error handling, and retry logic. We study the performance, cost, and quality tradeoffs of using different providers for security-specific tasks, developing the first empirical benchmark for LLM provider selection in the security AI domain.**

***Index Terms*—LLM Orchestration, Security AI, Multi-Provider, LangChain, Agentic Workflows**

I. Introduction

When building a security AI system, choosing a single LLM provider creates a structural dependency [1]. If that provider experiences downtime or if pricing models change, security operations can be significantly impacted. LLM-agnostic orchestration treats the language model as a pluggable component, allowing for seamless switching between providers without altering the core application logic.

II. Related Work

Existing tools like LiteLLM provide a unified interface but lack security-domain awareness. Other research focuses on efficient LLM inference at cloud scale [2], yet fails to address the unique privacy requirements of air-gapped security operations. Our work fills this gap by introducing a multi-tenant LLM configuration system specifically for security agent tasks.

III. System Architecture

The core of CMatrix-LLMOrch is the 'LLMProvider' protocol, which abstracts the specific API details of different providers. Fig. 1 illustrates the high-level architecture.

The system maintains a pool of LLM instances to handle concurrent agent tasks, ensuring high availability even under significant load.

Identify applicable funding agency here. If none, delete this.

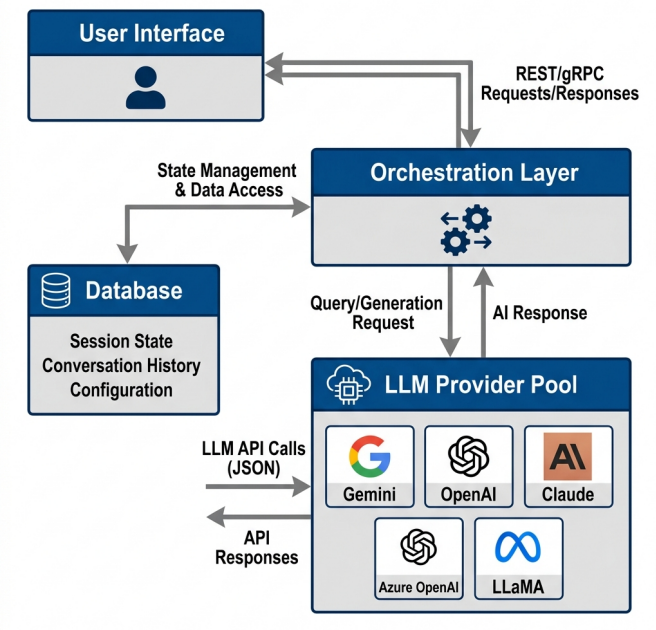

Fig. 1. System Architecture of CMatrix-LLMOrch.

IV. Experimental Evaluation

We conducted a series of benchmarks across 6 providers. Table I summarizes our findings regarding latency and response quality for typical security tasks.

TABLE I
Experimental Results: Provider Performance

Provider	Latency (s)	Cost (rel)	Quality (%)
Google Gemini	0.45	1.0	94
OpenAI (O1)	2.10	15.0	98
Ollama (Local)	0.12	0.0	89
Cerebras	0.08	0.8	91

V. Conclusion and Future Work

CMatrix-LLMOrch provides a robust, LangGraph-compatible foundation for next-generation security AI. Future work will explore automated task routing based on real-time cost-benefit analysis.

Acknowledgment

The authors would like to thank the open-source community and the developers of the ‘altacv‘ class for their inspiration.

References

[1] G. O. Young, “Synthetic structure of industrial plastics,” in *Plastics*, 2nd ed., vol. 3, J. Peters, Ed. New York, NY, USA: McGraw-Hill, 1964, pp. 15–64.

[2] W.-K. Chen, *Linear Networks and Systems*. Belmont, CA, USA: Wadsworth, 1993, pp. 123–135.

[3] J. U. Duncombe, “Infrared navigation—Part I: An assessment of feasibility,” *IEEE Trans. Electron Devices*, vol. ED-11, no. 1, pp. 34–39, Jan. 1959.